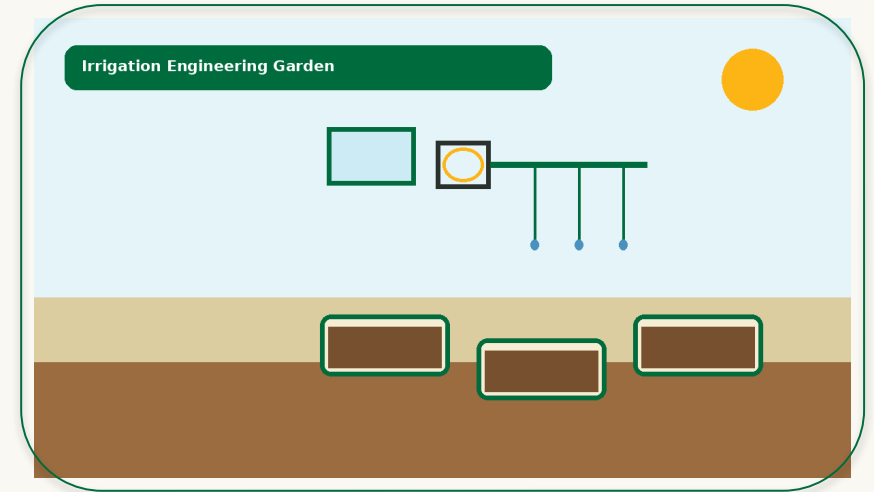


Irrigation Engineering Garden

Fundamentals of Irrigation Engineering

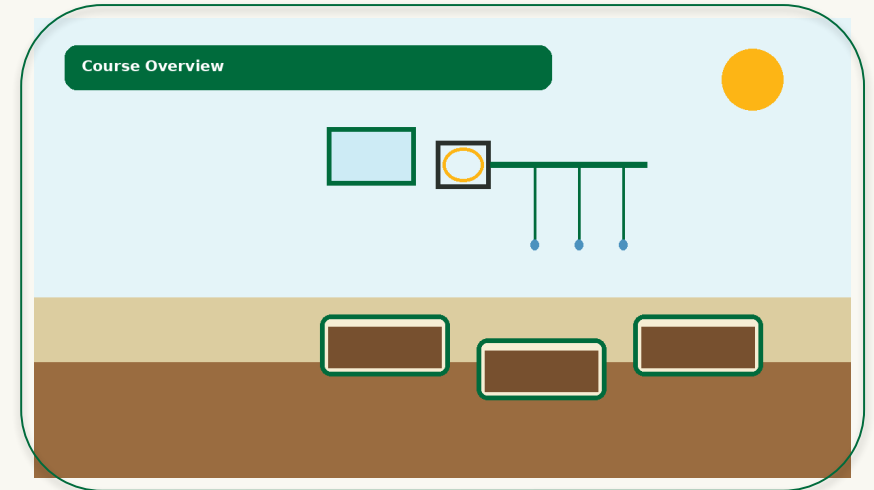
- LIFEWS AgriShine™ GrowMeal™ SSS 1-3 Teacher Training
- Audience: SSS 1-3 Agricultural Science, Biology, Mathematics and technical/STEM educators
- Format: Self-paced online course + supervised field/laboratory practical
- Recommended duration: 4 hours online + 2-3 hours practical
- SSS learning progression: design -> instrument -> model -> evaluate -> optimize -> innovate



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system at senior-secondary technical depth; teach all ten modules accurately; supervise design, instrumentation, measurement and experimental work; maintain defensible technical records; guide basic quantitative analysis; distinguish measured evidence from inference; connect results to management and engineering decisions.
- Core materials: teacher guide, system/plot drawings, crop and experiment plans, approved field tools, irrigation equipment, measuring tapes, scales, flow/pressure/moisture instruments where relevant, sensors/data sheets, clean harvest/sample containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe technical practical, and submit a production plan, engineering design, experimental protocol or analytical report appropriate to the course.



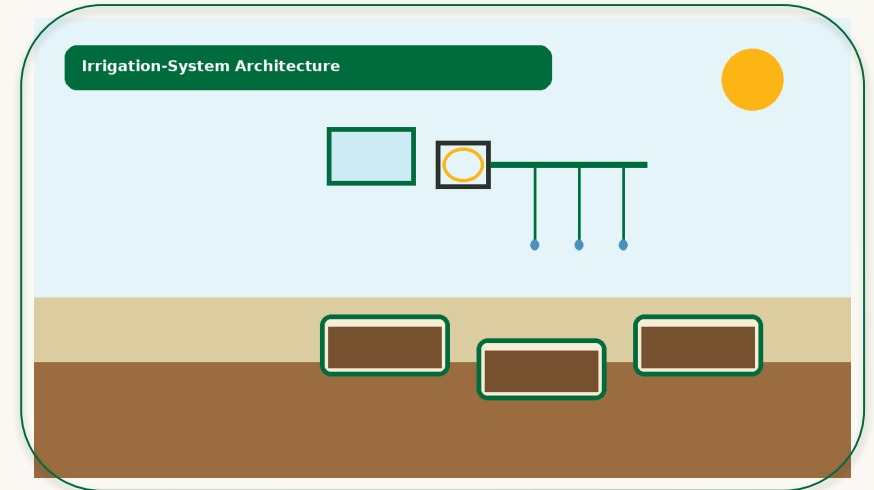
TEACHER PRACTICE

- Training sequence: DEFINE -> DESIGN -> IMPLEMENT -> MEASURE -> ANALYZE -> EVALUATE -> IMPROVE.

Irrigation-System Architecture

Teacher technical knowledge, laboratory/field practice and student supervision

- Map the system from source through storage/pump, filtration, control, conveyance, distribution and emitters to the crop root zone.
- Distinguish supply, control and application components.
- Identify isolation points and safe operating boundaries.
- Use a schematic with flow direction and component labels.
- Keep students away from mains electricity, open wiring and pressurized service operations.



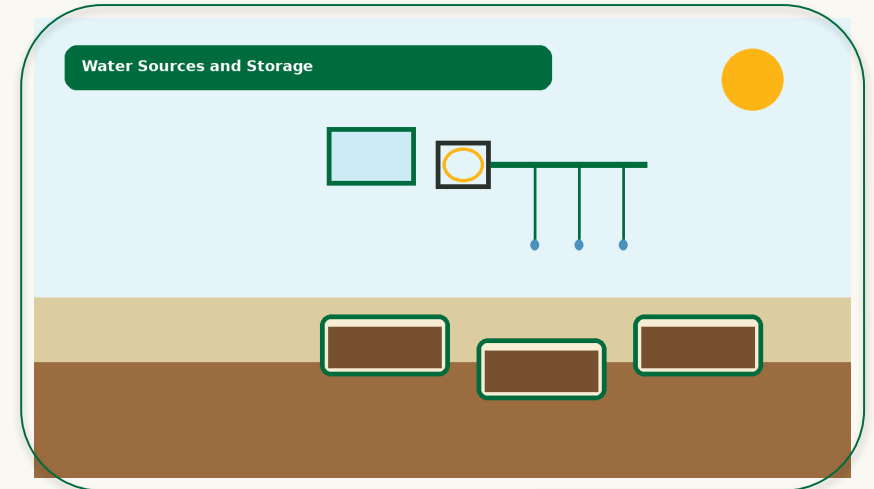
TEACHER PRACTICE

- Teacher task: produce a complete irrigation-system schematic.

Water Sources and Storage

Teacher technical knowledge, laboratory/field practice and student supervision

- Identify source type, available supply, water-quality considerations and storage capacity.
- Explain how storage can buffer differences between water availability and irrigation demand.
- Measure or estimate tank volume and monitor level.
- Include overflow, drainage, cover and structural safety in system assessment.
- Record source limitations that constrain irrigation design.



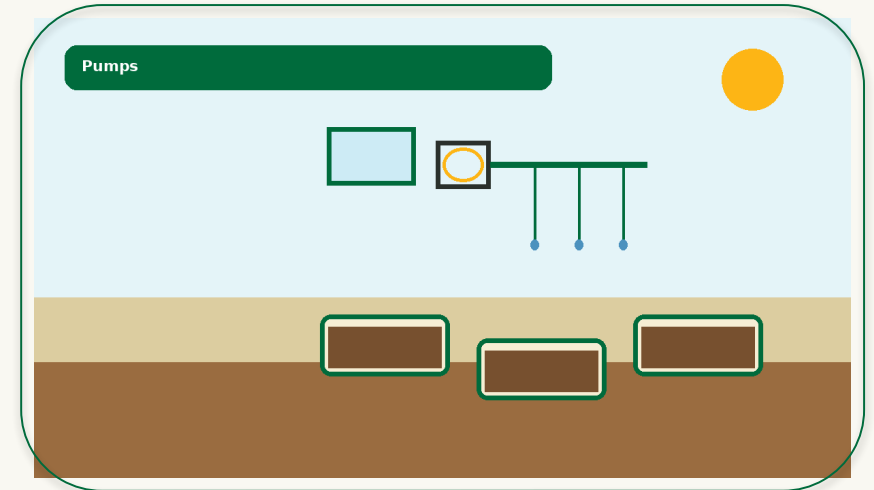
TEACHER PRACTICE

- Calculate usable storage volume from tank dimensions/specification.

Pumps

Teacher technical knowledge, laboratory/field practice and student supervision

- Explain pump function as adding energy to move water and/or create pressure.
- Identify pump type/nameplate information only from safe external labels.
- Relate flow, head/pressure and power conceptually.
- Do not permit students to open electrical enclosures, prime unsafe pumps or service rotating equipment.
- Compare pumped and gravity-fed sections where the school system includes both.



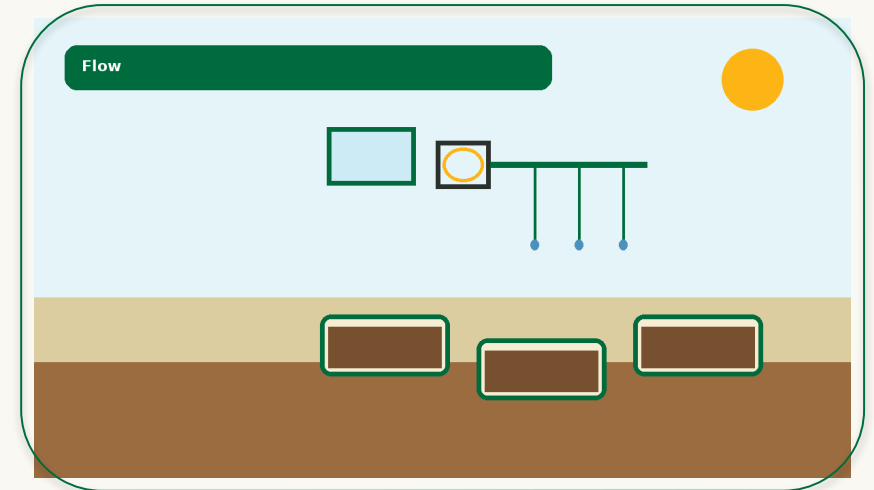
TEACHER PRACTICE

- Prepare a pump-function and safety briefing.

Flow

Teacher technical knowledge, laboratory/field practice and student supervision

- Define discharge/flow rate as volume per unit time.
- Measure using a flow meter or timed collection where safe.
- Record operating conditions during measurement.
- Use flow to calculate delivered volume over irrigation duration.
- Investigate major deviations for leaks, blockage, valve position or supply changes.



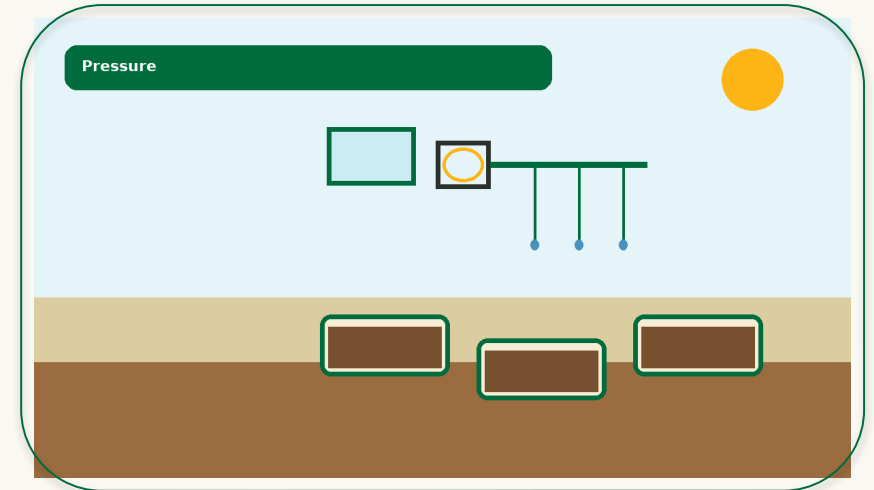
TEACHER PRACTICE

- Measure system flow and calculate hourly delivery.

Pressure

Teacher technical knowledge, laboratory/field practice and student supervision

- Define pressure as force per unit area and relate it to irrigation-system operation.
- Use installed gauges or approved low-pressure demonstration equipment.
- Record pressure at relevant system points.
- Explain pressure loss qualitatively along pipes and fittings.
- Never loosen pressurized fittings during student activities.



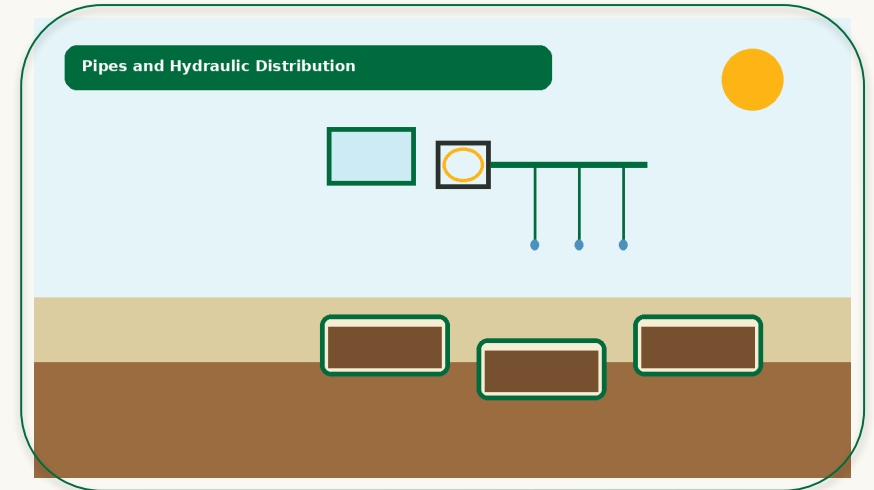
TEACHER PRACTICE

- Compare safe gauge readings at two system locations.

Pipes and Hydraulic Distribution

Teacher technical knowledge, laboratory/field practice and student supervision

- Identify mainline, submain/manifold and lateral functions.
- Relate pipe diameter, length, elevation and fittings to flow distribution qualitatively.
- Trace branch networks and zone boundaries.
- Inspect for leaks, kinks and physical damage.
- Use system drawings to connect physical layout with hydraulic behavior.



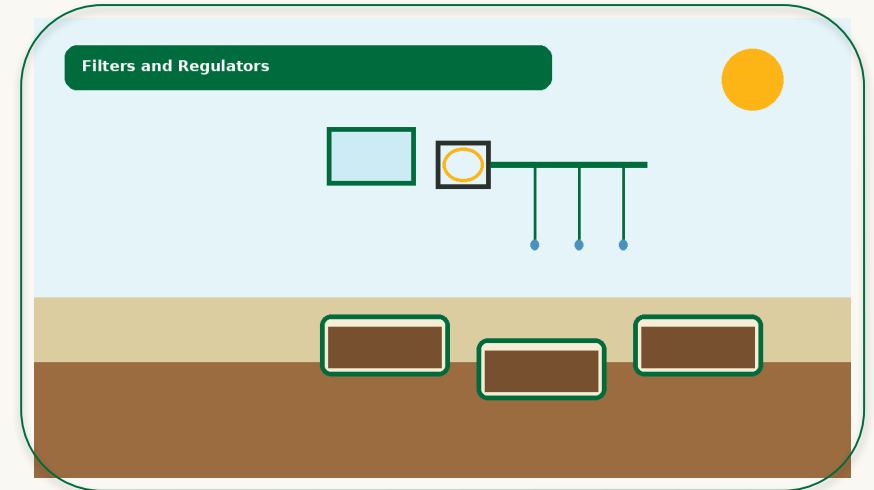
TEACHER PRACTICE

- Create a pipe-network map with flow direction.

Filters and Regulators

Teacher technical knowledge, laboratory/field practice and student supervision

- Explain filtration as protection against particles that can clog emitters.
- Explain pressure regulation as maintaining suitable downstream pressure where installed.
- Identify safe inspection indicators such as external gauges or clear housings.
- Adults perform disassembly and cleaning where tools or pressure isolation are required.
- Record maintenance intervals and observed problems.



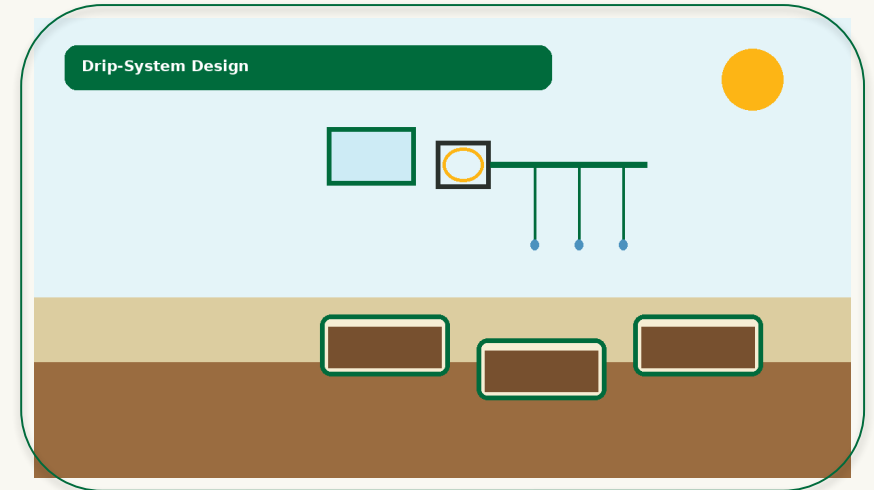
TEACHER PRACTICE

- Prepare a filter/regulator inspection checklist.

Drip-System Design

Teacher technical knowledge, laboratory/field practice and student supervision

- Select emitter spacing and lateral arrangement based on crop spacing and bed geometry.
- Estimate zone flow from number of emitters multiplied by nominal emitter discharge.
- Check whether source/pump capacity can support the planned zone.
- Include valves, filtration and flushing provisions.
- Design for maintainability and safe access, not only maximum coverage.



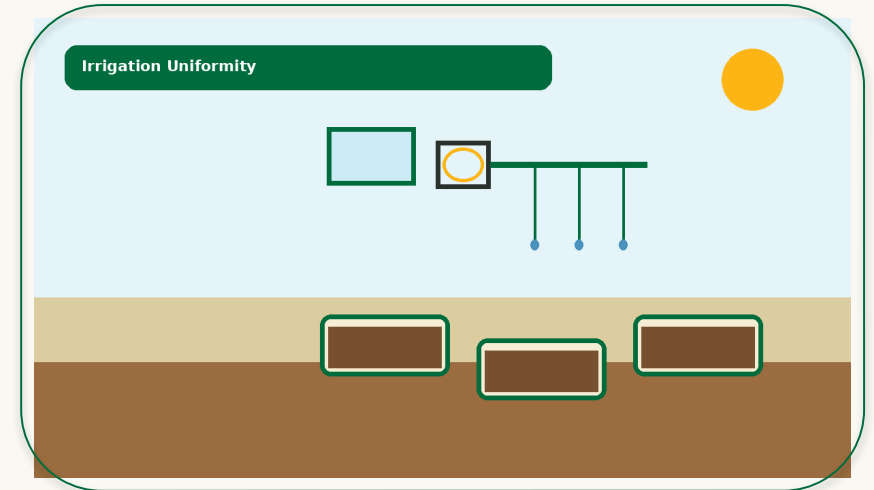
TEACHER PRACTICE

- Design one school-scale drip irrigation zone.

Irrigation Uniformity

Teacher technical knowledge, laboratory/field practice and student supervision

- Measure discharge from multiple emitters using equal collection times.
- Calculate simple average and compare low/high values or a basic uniformity indicator.
- Sample across beginning, middle and end of laterals.
- Identify whether poor uniformity may relate to pressure, blockage or design.
- Repeat measurements after maintenance to verify improvement.



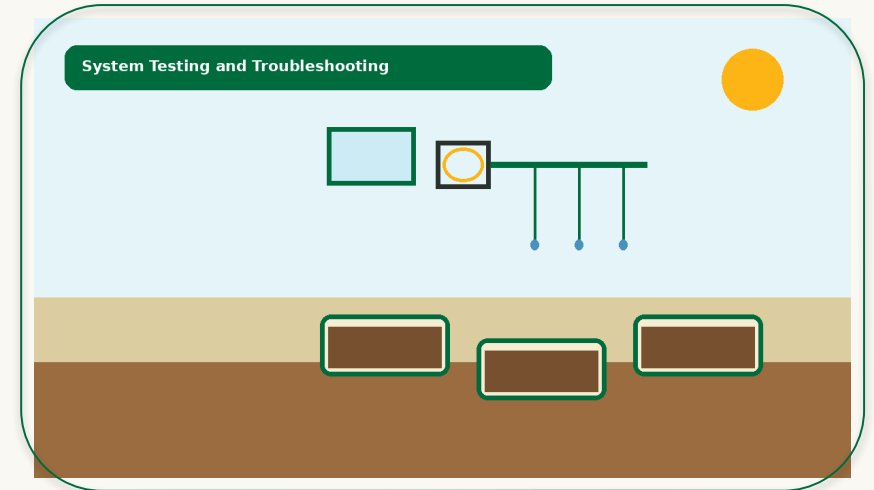
TEACHER PRACTICE

- Conduct an emitter-uniformity test.

System Testing and Troubleshooting

Teacher technical knowledge, laboratory/field practice and student supervision

- Use a structured sequence: inspect source/storage, check valves, observe pressure/flow, inspect leaks, test emitters, verify delivery.
- Change one condition at a time when diagnosing problems.
- Record symptom, test, finding and corrective action.
- Adults handle electrical, pump and pressurized repairs.
- Retest the system after correction.



TEACHER PRACTICE

- Completion task: develop and demonstrate an irrigation troubleshooting protocol.